

CS177 Bioinformatics: Software Development and Applications

Exercises on Bayesian Inference
Spring, 2025

Q1. Genetic Test

You are told that a genetic test is extremely good: 100% sensitive (that is, it is always correct if you have the disease) and 99.99% specific (that is, it gives a false positive only 0.01% of the time). A rare genetic disease of which only one in a million carriers is discovered. Using Bayes' theorem, explain why you might not want to take this genetic test.

Q1 Solution

- Let D denote the event of having the disease.
- Let e_D denote the event of test result of having the disease
- Then,

$$P(D) = 10^{-6}$$

$$P(\bar{D}) = 1 - 10^{-6} \approx 1.0$$

$$P(e_D|D) = 1.0$$

$$P(\bar{e}_D|\bar{D}) = 99.99\%$$

Q1 Solution (Continued)

- According to Bayes' Theorem:

$$\begin{aligned}P(D | e_D) &= \frac{P(e_D | D)P(D)}{P(e_D)} \\&= \frac{P(e_D | D)P(D)}{P(e_D | D)P(D) + P(e_D | \bar{D})P(\bar{D})} \\&\approx 0.01\end{aligned}$$

$$P(\bar{D} | e_D) \approx 99\%$$

Q2. Cancer diagnostic

A test for stomach cancer claims that it will report a positive result for 99.6% of people with the disease, and will report a negative result for 99.8% of those without the disease. It is known that the incidence rate of this disease is 7 per 100,000 in human population. If 600 people get positive results, how many of these people are expected to really have the disease? Justify your answer.

Q2 solution

Let the probability of positive test result given that the patient has the disease be denoted by $P(+result \mid + disease) = 0.996$. Likewise, the probability of negative test result given that the patient does not have the disease is $P(-result \mid - disease) = 0.998$.

It is known that the probability that the patient has the disease is $P(+disease) = 7 \times 10^{-5}$, and thus the probability that the patient does not have the disease is $P(-disease) = 1 - 7 \times 10^{-5} = 0.99993$.

Therefore, the probability of positive test result is

$$\begin{aligned} p(+test) &= p(+test \mid +disease) \times p(+disease) + p(+test \mid -disease) \times p(-disease) \\ &= p(+test \mid +disease) \times p(+disease) + [1 - p(-test \mid -disease)] \times p(-disease) \\ &= (0.996) \times (7 \times 10^{-5}) + (1.0 - 0.998) \times (0.99993) \\ &= 0.00206958. \end{aligned}$$

Q2 solution (continued)

By Bayes' theorem,

$$p(+disease \mid +test) = \frac{p(+test \mid +disease) \times p(+disease)}{p(+test)} = \frac{(0.996) \times 7 \times 10^{-5}}{0.00206958} = 0.03368799$$

Therefore, in the 600 people with positive test results, the expected number of people who really have the disease is $600 \times p(+disease \mid +test) = 20.21279 \approx 20$.